

SUMIT KUMAR SHARMA

AI ENGINEER

Bengaluru, Karnataka | +91 8273953988 | timusksharma@gmail.com

SUMMARY

AI Engineer with 5+ years of experience building production backend systems and, most recently, LLM-powered applications. Skilled in designing RAG pipelines and agentic workflows with LangChain and LangGraph, integrating OpenAI and Anthropic APIs, and serving models through FastAPI. Strong foundation in Python, distributed data processing with PySpark, and scalable API design — bringing an engineering-first approach to building reliable, maintainable AI systems.

SKILLS

AI / LLM	LangChain, LangGraph, RAG pipelines, embeddings, OpenAI & Anthropic APIs, Deepgram, vector DBs (Pinecone, Qdrant, Milvus, FAISS), prompt engineering
ML / Data Science	PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, pandas, NumPy, PySpark
Backend	Python, FastAPI, Node.js, Express.js, REST APIs, Microservices, OAuth 2.0
Databases	PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch
MLOps & Tools	Docker, Kubernetes, MLflow, model deployment, Git, Jira, Kafka
Frontend	React.js, JavaScript (ES6+)

EXPERIENCE

Senior Software Engineer — Alten India (Client: Airbus)

Bengaluru, Karnataka

Aug 2022 – Present

- Built an Aircraft Component Intelligence & Compliance Assistant combining PySpark processing of structured aircraft/component datasets with RAG over technical documentation — letting users query component data in natural language, run technical/compliance analysis, and generate automated reports.
- Developed an AI-powered Dataset Health & Observability Platform that monitors dataset freshness, build status, schema changes, primary-key/duplicate issues, and pipeline health, with AI-assisted root-cause analysis and remediation recommendations.
- Integrated LLMs, RAG, vector search, and MCP-based tooling into enterprise data workflows, enabling AI systems to retrieve engineering information, investigate dataset/component issues, and deliver context-aware responses bridging structured Skywise data with unstructured documentation.
- Designed FastAPI backend APIs to expose PySpark-processed datasets, enabling downstream applications and users to query and consume structured Airbus data through scalable REST services.
- Developed scalable PySpark, Python, and SQL data pipelines for Airbus, processing millions of records per run across aircraft, component, manufacturing, incident, programme, and operational datasets.
- Built and maintained Raw → Restricted → Final data products on Palantir Foundry — covering ingestion, transformation, data-quality validation, lineage, publishing, and access management.
- Optimized large-scale PySpark workloads with partitioning, caching, partition pruning, and broadcast joins, improving pipeline execution efficiency by ~30–40%.
- Automated pipeline scheduling, dataset refresh, monitoring, and failure handling to improve production reliability and meet data-pipeline SLAs, supporting Airbus manufacturing, programme, incident, facility-management, and maintenance KPI use cases.

Software Developer — Salesken.ai

Bengaluru, Karnataka

Nov 2021 – Aug 2022

- Developed AI-powered conversation-intelligence features for Salesken's customer-engagement and CRM platform, giving enterprise sales and customer-care teams real-time insights, contextual recommendations, and live coaching during customer conversations.
- Contributed to the real-time conversational-AI pipeline — processing live Twilio calls through Deepgram speech-to-text, semantic analysis, embeddings, Milvus vector search, and RAG to detect customer intent, pricing/budget objections, and context, delivering actionable recommendations to reps over WebSockets.

- Built scalable backend services in Python and FastAPI for CRM workflows, call management, conversation intelligence, real-time notifications, and AI-powered customer-interaction features.
- Developed event-driven processing with Kafka for asynchronous handling of call events and conversation data, using AWS S3 for scalable storage of recordings and conversation artifacts.
- Contributed to post-call analytics, mining transcripts and recordings for keywords, sentiment, objections, rep performance, and conversion outcomes to explain successful and unsuccessful interactions.
- Implemented OAuth 2.0 Salesforce and HubSpot integrations to automate sync of customer, lead, task, and pipeline data, and optimized production APIs for performance, reliability, and scalability while collaborating cross-functionally in Agile.

Software Engineer — Oracle Cerner

Sep 2021 – Nov 2021

Bengaluru, Karnataka

- Developed and maintained enterprise healthcare web applications using JavaScript and Node.js, contributing to core features and business workflows.
- Built and enhanced backend REST APIs, partnering with senior engineers to ship functionality and resolve production issues to enterprise standards.
- Contributed to code reviews, testing, bug fixing, and Agile sprint activities in a collaborative development environment.

PROJECTS

AI Data Pipeline Generator — LLM Agent • PySpark • ETL

Stack: Python, FastAPI, LangGraph, PySpark, SQL, OpenAI, Docker, Kubernetes

- Built an AI-powered agent that generates production-ready ETL pipelines from natural-language specifications, automating SQL generation, PySpark transformations, schema validation, and data-quality checks.
- Generated configurable pipeline templates with logging, error handling, and deployment-ready project structures, cutting manual effort for repetitive ETL workflows through intelligent code generation.

Enterprise AI Knowledge Assistant — RAG • Vector Search

Stack: Python, FastAPI, LangGraph, LangChain, OpenAI/Claude, Qdrant, Redis, PostgreSQL, Docker, Kubernetes

- Built an enterprise-grade RAG platform answering questions over private documents, with an end-to-end ingestion pipeline supporting PDF, DOCX, Excel, HTML, and Markdown files with automated chunking and metadata extraction.
- Implemented hybrid semantic search (vector embeddings, metadata filtering, reranking) and streaming FastAPI endpoints, deployed on Kubernetes with observability and monitoring.

Multi-Agent Workflow Automation Platform — LangGraph • MCP

Stack: Python, LangGraph, MCP, FastAPI, Redis, PostgreSQL, Docker

- Developed a multi-agent platform where Planner, Researcher, Executor, and Reviewer agents collaborate to solve complex business tasks, with tool calling, long-term memory, checkpointing, and human-in-the-loop approvals.
- Built reusable MCP tools for database operations, web search, and document processing, with resilient workflows featuring automatic retries and state persistence.

AI Frontend Engineering Agent — Figma-to-Code • Agentic

Stack: Python, LangGraph, MCP (Figma API), OpenAI/Claude, React, TypeScript, FastAPI, Docker

- Built an agentic system that ingests Figma designs via MCP, converts them into a structured UI representation, and generates production-ready React/TypeScript code.
- Implemented an iterative validate-and-refine loop where the agent verifies the generated application and modifies code from natural-language instructions.

EDUCATION

B.Tech, Computer Science and Engineering

2017 – 2021

Lovely Professional University, Jalandhar, Punjab | Languages: English, Hindi